

FRANK PORTER
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M1 MACRO

Group A

Session 13

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Dynamic Labor Demand model

$$Y = F(L) = \alpha_0 L_t - \frac{\alpha_1}{2} L_t^2$$

$$F'(L) = \alpha_0 - \alpha_1 L_t > 0 \text{ because } \alpha_0 \text{ is "large enough"}$$

$$F'' = -\alpha_1 < 0$$

Ads.

$$\bullet \text{ Cost fnct: } C(\Delta L_t) = \frac{b}{2} (L_t - L_{t-1})^2$$

$$\Delta L_t = L_t - L_{t-1}$$

$$\bullet W_t = \text{wage}$$

Firm problem

- Price of production = 1

$$\text{Max}_{\{L_{t+i}\}} V_t = \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} \Pi_{t+i}$$

$$\Pi_{t+i} = \text{profit} = F(L_{t+i}) - W_{t+i} L_{t+i} - c(L_{t+i} - L_{t+i-1})$$

$$V_t = \sum_{i=0}^{\infty} \left(\frac{1}{1+r} \right)^i \times \left[\alpha_0 L_{t+i} - \frac{\alpha_1}{2} L_{t+i}^2 - W_{t+i} L_{t+i} - \frac{b}{2} (L_{t+i} - L_{t+i-1})^2 \right]$$

$$\text{Max}_{\{L_{t+i}\}_{i=0}^{\infty}} V_t = \sum_{i=0}^{\infty} \left(\frac{1}{1+r}\right)^i \left(\alpha_0 \underline{L_{t+i}} - \frac{\alpha_1}{2} \underline{L_{t+i}}^2 - W_{t+i} \underline{L_{t+i}} - \frac{b}{2} (\underline{L_{t+i}} - \underline{L_{t+i-1}})^2 \right)$$

$$\text{foc } L_{t+i} / : \left(\frac{1}{1+r}\right)^i (\alpha_0 - \alpha_1 L_{t+i} - W_{t+i} - b(L_{t+i} - L_{t+i-1}))$$

$$+ \left(\frac{1}{1+r}\right)^{i+1} b (L_{t+i+1} - L_{t+i}) = 0$$

→ I look at $i=0$ (for convenience) $\forall i \geq 0$

FOC

$$d_0 - d_1 L_t - W_t - b(l_t - l_{t-1}) + \underbrace{\frac{1}{1+\alpha}}_{=\beta} b(l_{t+1} - l_t) = 0$$

If $b=0$: No adjustment cost

$$d_0 - d_1 L_t - W_t = 0$$

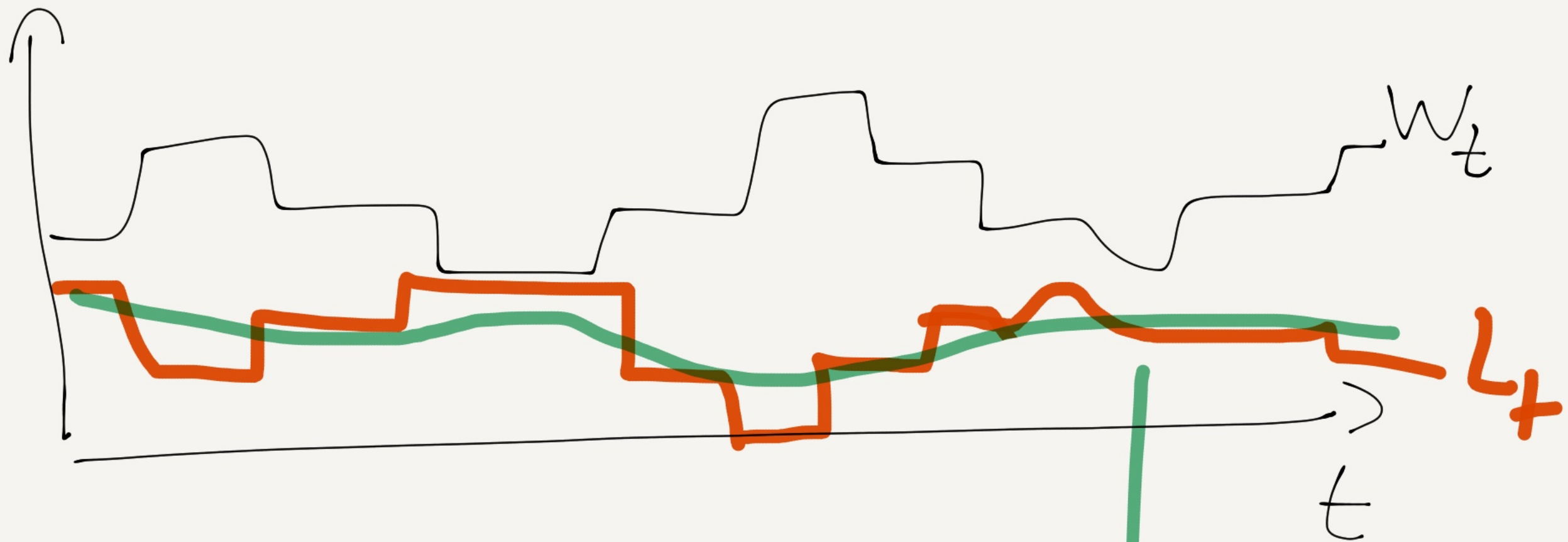
optimal
choice

optimal size

$$L_t = \frac{d_0 - W_t}{d_1}$$

(assuming $d_0 - W_t > 0$)

STATIC CHOICE



with adjusted
coords ($b \neq 0$)

. When $b \neq 0$

$$(1) \quad d_0 - d_1 L_t - W_t - b(L_t - L_{t-1}) + \rho b(L_{t+1} - L_t) = 0$$

Solving the model = finding L_t at every $t \geq 0$
given L_{-1}
such that (1) is satisfied.

Math Property

$$aX_{t+1} + bX_t + cX_{t-1} = 0 \quad (*)$$

characteristic polynomial of this equation.

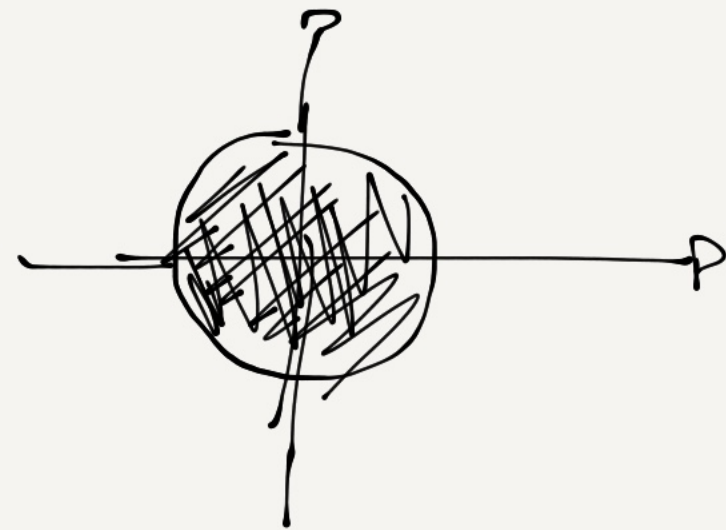
$$P(\lambda) = a\lambda^2 + b\lambda + c = 0$$

let λ_1, λ_2 be the two roots of that polynomial
(real or complex)

$$\hookrightarrow \text{meaning: } \begin{aligned} P(\lambda_1) &= 0 \\ P(\lambda_2) &= 0 \end{aligned}$$

The solution of (*) writes

$$X_t = c_1 \lambda_1^t + c_2 \lambda_2^t$$



where c_1 and c_2 are determined by
2 boundary conditions.

for ex: if ~~X_0~~ , ~~X_1~~ are given, then

2 eq in (c_1, c_2)

$$\begin{cases} X_0 = c_1 \lambda_1^0 + c_2 \lambda_2^0 = c_1 + c_2 \\ X_1 = c_1 \lambda_1 + c_2 \lambda_2 \end{cases}$$

$$(1) \alpha_0 - \alpha_1 l_t - w_t - b(l_t - l_{t-1}) + \beta b(l_{t+1} - l_t) = 0$$

* Steady State

• Assume $w_t = \bar{w} \quad \forall t$

• $\bar{L}$ is the solution of

$$\alpha_0 - \alpha_1 \bar{L} - \bar{w} - b(\bar{L} - \bar{L}) + \beta b(\bar{L} - \bar{L}) = 0$$

↳ (1) when $l_t = \bar{L} \quad \forall t$

$$\boxed{\bar{L} = \frac{\alpha_0 - \bar{w}}{\alpha_1}}$$

Define $\hat{L}_t = L_t - \bar{L}$

$$(1) \quad \alpha_0 - \alpha_1 \left(\overbrace{L_t - \bar{L} + \bar{L}}^{\hat{L}_t} \right) \cdot \left(\overbrace{w_t - \bar{w}}^{\hat{w}_t} \right) - b \left(\overbrace{L_t - \bar{L} + \bar{L}}^{\hat{L}_t} - \overbrace{L_{t-1} - \bar{L} + \bar{L}}^{\hat{L}_{t-1}} \right) \\ + \beta b \left(\overbrace{L_{t+1} - \bar{L} + \bar{L}}^{\hat{L}_{t+1}} - \overbrace{L_t - \bar{L} + \bar{L}}^{\hat{L}_t} \right) = 0$$

(=)

$$\alpha_0 - \alpha_1 \hat{L}_t - \alpha_1 \bar{L} - \hat{w}_t - b(\hat{L}_t - \hat{L}_{t-1}) - b(\bar{L} - \bar{L}) \\ + \beta b(\hat{L}_{t+1} - \hat{L}_t) + \beta b(\bar{L} - \bar{L}) = 0$$

$$\alpha_0 - \alpha_1 \bar{L} - \bar{w} = \alpha_0 - \alpha_1 \left(\frac{\alpha_0 - \bar{w}}{\alpha_1} \right) - \bar{w} = 0$$

therefore

$$(1) \Rightarrow -\alpha, \hat{L}_t - \hat{w}_t - b(\hat{L}_t - \hat{L}_{t-1}) + \beta b(\hat{L}_{t+1} - \hat{L}_t) = 0$$

$$\Leftrightarrow \boxed{\beta b \hat{L}_{t+1} - (d_1 + b + \beta b) \hat{L}_t + b \hat{L}_{t-1} = \hat{w}_t}$$

Assume $\hat{w}_t = 0 \forall t \Rightarrow w_t = \overline{w}$

$$\boxed{\beta b \hat{L}_{t+1} - (d_1 + b + \beta b) \hat{L}_t + b \hat{L}_{t-1} = 0}$$

Solve $\beta b \Sigma_{tT_1} - (\alpha_1 + b + \beta b) \Sigma_t + b \Sigma_{t-1} = 0$

Characteristic polynomial:

$$P(\lambda) = \beta b \lambda^2 - (\alpha_1 + b + \beta b) \lambda + b = 0$$

λ_1, λ_2 are the two roots

Properties of polynomial if $P(\lambda) = a\lambda^2 + b\lambda + c$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_1 + \lambda_2 = -\frac{b}{a}$$

$$, \lambda_1 \times \lambda_2 = \frac{c}{a}$$

Here $\lambda_1 + \lambda_2 = \frac{1}{\beta} > 1$ (because $\beta = \frac{1}{1+\alpha} < 1$)

$$\lambda_1 + \lambda_2 = \frac{a_1 + b + \beta b}{\beta b} > 0$$

$\lambda_1 \lambda_2 > 0 \Rightarrow \lambda_1$ and λ_2 have the same sign

$\lambda_1 + \lambda_2 > 0 \Rightarrow \lambda_1$ and λ_2 are both positive

$\lambda_1 \lambda_2 > 1 \Rightarrow \lambda_1$ ^{and/or} λ_2 is > 1

$\rightarrow$ say it's λ_2

$P(1) = -\alpha < 0$

$P(\lambda)$? $\checkmark \rightarrow$ because $\beta b > 0$



$\Rightarrow$ proves that $\lambda_1 \in \boxed{0, 1}$

Therefore, the solution is

$$\hat{L}_t = c_1 \lambda_1^t + c_2 \lambda_2^t$$

for some c_1, c_2 to be found.

